MOULIK JAIN | DS 402

Graph-Based Financial Intelligence: Detecting Money Laundering with GNNs

Motivation

$2 Trillion Laundered Annually, Detection is Broken

- Traditional rule-based systems produce 95%+ false positive rates, analysts are overwhelmed
- Each transaction looks normal in isolation; **the fraud is in the pattern across accounts**

Two Real-World Benchmarks

IBM AML HI-Medium

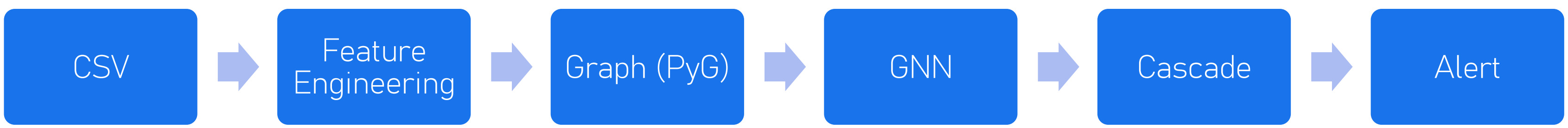
- 2M transactions, ~500k accounts
- 0.003% fraud rate
- 8 laundering pattern types (fan-out, cycle, stack…)
- Edge-level classification

Elliptic Bitcoin

- 203k nodes, 234k edges
- 9.76% fraud rate
- Cross-domain generalization test

60/20/20 stratified split · Focal Loss for class imbalance · V100 GPU on PSC Bridges2

Pipeline: Raw Transactions → Alert

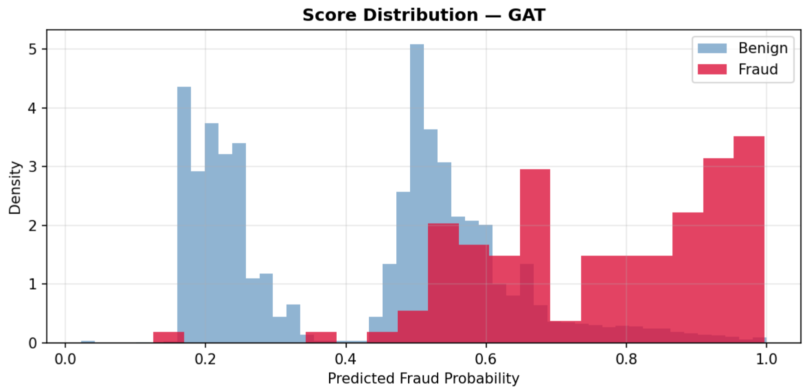


Features: 16 node feats (PageRank, k-core, degree stats) + 11 edge feats (structuring flags, delta-t, amount)

Models tested: GraphSAGE · GAT · TGN · H2GCN

Why Standard GNNs Struggle and How I Fixed It

- Fraudsters deliberately route money through legitimate accounts
- Standard message passing **averages** neighbors → fraud signal gets washed out
- **H2GCN fix:** keep ego-embedding separate; concatenate [self + 1-hop + 2-hop] instead of averaging



GAT assigns high fraud scores to tens of thousands of benign transactions, precision collapses at any threshold

Graph Structure Beats Tabular — H2GCN Leads

Model	Recall	ROC-AUC	PR-AUC
RF (no graph)	0.718	0.977	0.152
XGBoost (no graph)	0.581	0.972	0.511
GraphSAGE	0.798	0.947	0.042
GAT	0.960	0.902	0.005
TGN	0.623	0.974	0.136
H2GCN	**0.903**	**0.960**	**0.113**

H2GCN achieves the best ROC-AUC of any GNN and 22× better PR-AUC than GAT

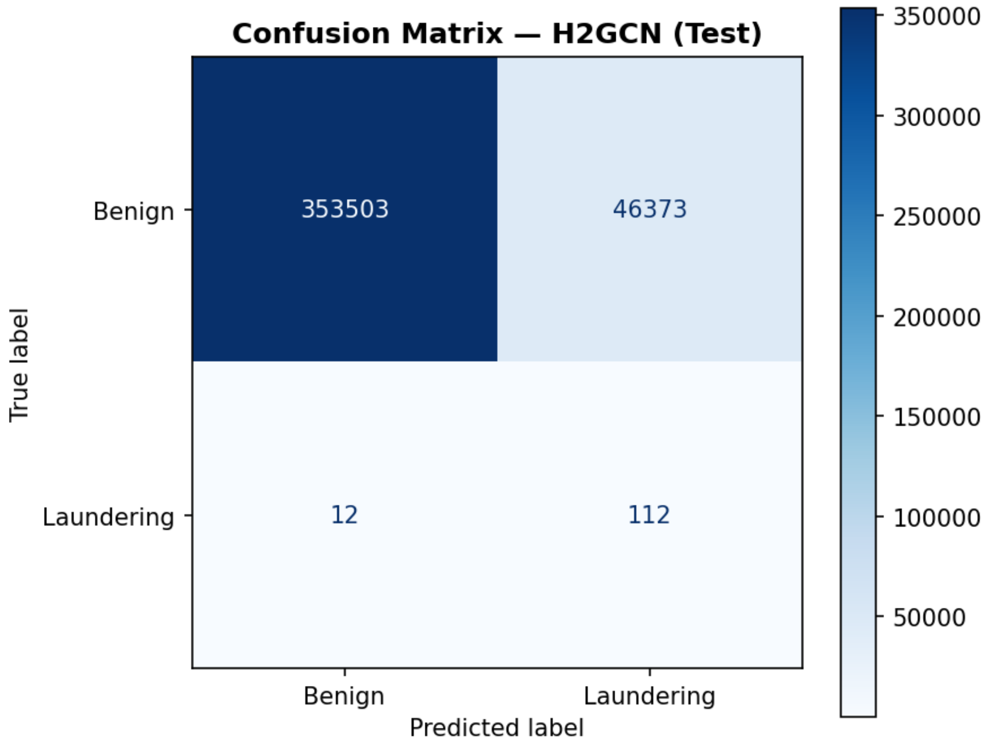
High Recall Alone Is Not Deployable

The problem (one stark stat):

- GAT flags 96% of fraud.
- Precision = **0.06%.**
- Every real fraud alert comes with **1,667 false alarms.**

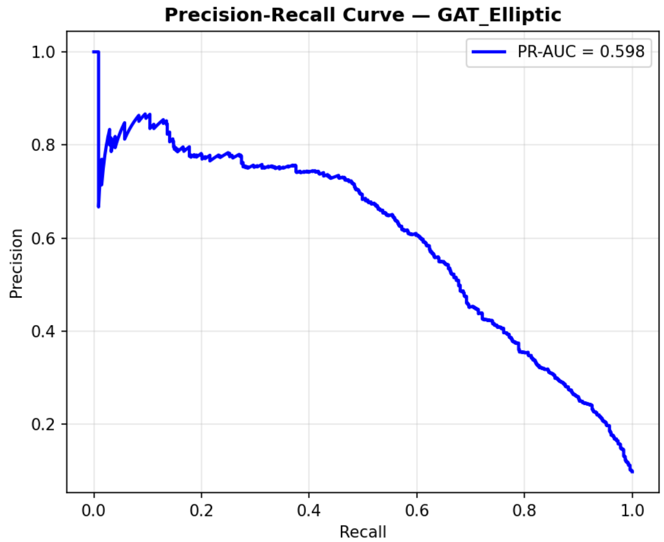
The solution table:

	H2GCN	H2GCN + Cascade
Recall	90.3%	54.8%
Precision	0.24%	82.9%
F1	0.005	0.660
PR-AUC	0.113	0.595
ROC-AUC	0.960	0.993



46,373 false positives, still a problem the cascade solves

Generalises to Elliptic Bitcoin



GAT on Elliptic: 94.4% recall · PR-AUC = 0.599 · ROC-AUC = 0.906

Different domain, different asset class, different fraud rate; the GNN approach transfers

Conclusion & Future Work

Key Takeaways
Graph structure is necessary — GNNs consistently beat tabular baselines on recall
Heterophily is the core challenge — H2GCN's ego-separation gives 22× better PR-AUC than GAT
The two-stage cascade (GNN → XGBoost) is the right production design: F1 = 0.66, precision = 82.9%
Future work
Scale to full 13M-edge dataset with heterogeneous bank-account graph
Deploy TGN memory module for real-time streaming detection

Citations: Weber et al. 2019 · Zhu et al. 2020 (H2GCN) · Hamilton et al. 2017 (GraphSAGE) · Lin et al. 2017 (Focal Loss) · Rossi et al. 2020 (TGN)

Graph structure is not just useful for AML, it is essential.

THANKS!